

DESCRIPTIVE STATISTICS

Complete Detailed Notes for Statistics, Data Science and Machine Learning Basics

What this PDF covers	Statistics basics, data types, central tendency, position, dispersion, skewness, kurtosis, formulas, examples, and interview revision.
Best for	CDAC, data analytics, data science, machine learning, statistics exams, and interview preparation.
Study method	Learn concept -> memorize formula -> solve example -> revise summary tables.

Golden Rule: If data has outliers, prefer Median, IQR and Quartile Deviation. If data is clean and symmetric, Mean and Standard Deviation work well.

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1. Statistics and Descriptive Statistics

Statistics is a branch of mathematics that deals with collecting, organizing, presenting, analyzing and interpreting data. In data science, statistics helps us understand patterns, trends, variation, uncertainty and relationships in data before building models.

Why statistics is important in data science

- **Data cleaning:** identify irrelevant data, duplicates, missing values and outliers.
- **Data analysis:** summarize data and find useful patterns before modeling.
- **Understanding data:** calculate mean, median, variance, standard deviation, probability and distributions.
- **Decision making:** convert raw data into useful information for reports and predictions.

Descriptive vs Inferential Statistics

Type	Meaning	Example
Descriptive Statistics	Describes and summarizes the available data only.	Average marks of 50 students, highest and lowest marks, standard deviation.
Inferential Statistics	Uses a sample to make conclusions about a larger population.	Survey 100 students and estimate opinion of the whole college.

Descriptive statistics is usually the first step in data analysis. It does not generalize beyond the given data. Inferential statistics includes hypothesis testing, confidence intervals and estimation of parameters.

2. Types of Data

2.1 Based on source

Type	Meaning	Example
Primary Data	Data collected directly from the field or individuals.	Survey, observation, interview, experiment, teacher measuring student heights.
Secondary Data	Data already collected by someone else.	Government reports, Kaggle datasets, company records, medical records.

2.2 Based on number of variables

Type	Meaning	Example
Univariate	Only one variable is studied.	Height of students, temperature of Delhi for 7 days.
Bivariate	Two variables are studied together.	Price and demand, distance and time, study hours and marks.
Multivariate	More than two variables are studied together.	Profit depends on sales, cost, labour, marketing expense etc.

2.3 Based on organization

Type	Meaning
Raw Data	Unorganized data collected initially. Example: 3, 104, 46, 45, 43, 60.
Organized Data	Data arranged in tables, categories, class intervals or frequency distributions.

2.4 Based on values

Type	Meaning	Examples
Discrete Data	Countable values, usually whole numbers.	Number of patients, phone calls, rooms, defective items.
Continuous Data	Measurable values that can take decimals within a range.	Height, weight, temperature, blood pressure, waiting time.

3. Stages in Statistical Investigation

A statistical investigation normally follows these five stages:

- Collection of data
- Organization of data
- Presentation of data using tables or graphs
- Analysis of data
- Inference or conclusion from the data

After collection, data may be organized as raw data, ungrouped frequency distribution for discrete data, or grouped frequency distribution for continuous data. It can then be presented using bar charts, pie charts, pictographs, histograms and frequency polygons.

4. Measures of Central Tendency

Measures of central tendency identify the central or representative value of a dataset. The main measures are Mean, Median and Mode.

Measure	Meaning	Best used when
Mean	Arithmetic average.	Numerical data is clean and symmetric with no outliers.
Median	Middle value after sorting.	Data is skewed or contains outliers.
Mode	Most frequently occurring value.	Categorical data or most common value is needed.

4.1 Arithmetic Mean

The arithmetic mean is the average value of the dataset. It is the most commonly used measure of central tendency but it is sensitive to outliers.

$$\text{For raw data: } \bar{x} = (\text{sum } x) / N$$

Example: For data 10, 20, 30, 40:

$$\bar{x} = (10 + 20 + 30 + 40) / 4 = 25$$

Mean for frequency distribution

$$\bar{x} = (\sum f x) / (\sum f)$$

X	f	fx
2	2	4
3	1	3
7	3	21
8	1	8
Total	7	36

$$\bar{x} = 36 / 7 = 5.15$$

4.2 Shortcut Method for Mean

When values are large, choose an assumed mean A and calculate deviations from A.

$$\bar{x} = A + (\sum f d) / N, \text{ where } d = x - A$$

4.3 Step Deviation Method

Useful for grouped data with equal class interval.

$$\bar{x} = A + [(\sum f u) / (\sum f)] \times i, \text{ where } u = (x - A) / i$$

For continuous grouped data, first calculate class marks:

$$\text{Class mark} = (\text{lower limit} + \text{upper limit}) / 2$$

4.4 Geometric Mean

Geometric mean is useful for growth rates, ratios, index numbers and compound returns.

$$GM = \text{nth root of } (x_1 \times x_2 \times \dots \times x_n)$$

4.5 Harmonic Mean

Harmonic mean is useful for averaging rates, especially average speed when distance is same.

$$HM = N / [(1/x_1) + (1/x_2) + \dots + (1/x_n)]$$

Example: A cyclist goes from home to college at 10 km/hr and returns at 15 km/hr. Since distance is same, use harmonic mean.

$$HM = 2 / [(1/10) + (1/15)] = 12 \text{ km/hr}$$

4.6 Weighted Mean

Weighted mean is used when every value does not have equal importance.

$$\text{Weighted mean} = (\sum w x) / (\sum w)$$

Subject	Marks	Weight	Marks x Weight
English	60	1	60
Biology	75	2	150
Mathematics	63	1	63
Physics	59	3	177
Chemistry	55	3	165

Subject	Marks	Weight	Marks x Weight
Total		10	615

$$\text{Weighted mean} = 615 / 10 = 61.5$$

4.7 Trimmed Mean

A trimmed mean is calculated after removing a fixed percentage of smallest and largest values from the dataset. It reduces the effect of outliers.

Example: In data 10, 20, 30, 40, 500, the value 500 is an outlier. A trimmed mean gives a more stable central value than ordinary mean.

4.8 Median

Median is the middle value when data is arranged in ascending order. It divides the data into two equal halves and is robust to outliers.

$$\text{If } N \text{ is odd: Median} = [(N + 1) / 2] \text{th term}$$

$$\text{If } N \text{ is even: Median} = \text{average of } (N/2) \text{th and } [(N/2) + 1] \text{th terms}$$

Example: Data 5, 8, 10, 15, 20 has $N = 5$. Median = 3rd term = 10.

Example: Data 5, 8, 10, 15 has $N = 4$. Median = $(2\text{nd} + 3\text{rd terms})/2 = (8 + 10)/2 = 9$.

Median for grouped data

$$\text{Median} = l + [(N/2 - C) / f] \times h$$

Symbol	Meaning
l	Lower boundary of the median class
N	Total frequency
C	Cumulative frequency before median class
f	Frequency of median class
h	Class width

The median class is the class whose cumulative frequency is just greater than $N/2$.

4.9 Mode

Mode is the value that occurs most frequently. It is very useful for categorical data.

Data	Mode
5, 3, 5, 8, 9	5
8, 9, 9, 7, 8, 2, 5	8 and 9 (bimodal)
4, 12, 3, 6, 7	No mode

Mode for grouped data

$$\text{Mode} = l + [(f_1 - f_0) / (2f_1 - f_0 - f_2)] \times h$$

Symbol	Meaning
l	Lower limit of modal class

Symbol	Meaning
f1	Frequency of modal class
f0	Frequency of preceding class
f2	Frequency of succeeding class
h	Class size

The class interval with maximum frequency is the modal class.

For unimodal data: Mode = 3 Median - 2 Mean

4.10 Missing Value Imputation

Method	Use for	Why
Mean	Numerical, symmetric data	Works well if there are no outliers.
Median	Numerical, skewed data or outliers	Robust against extreme values.
Mode	Categorical data	Matches an existing category.

Data science rule: Salary, income and house price often have outliers, so median is usually safer than mean. For gender, city or product category, use mode.

5. Measures of Position: Quantiles

Measures of position identify the position of a value relative to other values in a dataset. Quartiles, deciles and percentiles are collectively called quantiles.

5.1 Quartiles

Quartiles divide an ordered dataset into four equal parts.

Measure	Meaning
Q1	First quartile. 25% of data is less than or equal to it.
Q2	Second quartile. 50% of data is less than or equal to it. It is the median.
Q3	Third quartile. 75% of data is less than or equal to it.

5.2 Deciles and Percentiles

Measure	Meaning
Deciles	Divide data into 10 equal parts. D5 is the median.
Percentiles	Divide data into 100 equal parts. P50 is the median.

Q2 = D5 = P50 = Median

5.3 Position formulas for raw data

Q1 position = $(N + 1) / 4$

Q3 position = $3(N + 1) / 4$

Dk position = $k(N + 1) / 10$

Pk position = $k(N + 1) / 100$

For individual and discrete data, use $N + 1$ in position formulas. For continuous grouped data, use N instead of $N + 1$.

5.4 Quartile example

Find Q_1 , Q_2 and Q_3 for data: 5, 7, 9, 12, 15, 18, 20, 22, 25.

$N = 9$

$$Q_1 \text{ position} = (9 + 1)/4 = 2.5\text{th term}$$

2.5th term lies halfway between 2nd term 7 and 3rd term 9.

$$Q_1 = 7 + 0.5(9 - 7) = 8$$

$$Q_2 = 2(9 + 1)/4 = 5\text{th term} = 15$$

$$Q_3 = 3(9 + 1)/4 = 7.5\text{th term} = 20 + 0.5(22 - 20) = 21$$

6. Measures of Dispersion / Spread

Dispersion means the degree to which data values spread around a central value such as mean or median. Less dispersion means data is more consistent.

Measure	Meaning	Outlier sensitivity
Range	Maximum - Minimum	Very high
IQR	$Q_3 - Q_1$, spread of middle 50%	Low
Quartile Deviation	$(Q_3 - Q_1)/2$	Low
Mean Deviation	Average absolute deviation from mean/median	Moderate
Variance	Average squared deviation from mean	High
Standard Deviation	Square root of variance	High
Coefficient of Variation	$SD/\text{Mean} \times 100$	Used for comparison

6.1 Range

$$\text{Range} = \text{Maximum value} - \text{Minimum value}$$

Range is simple but crude because it depends only on two extreme values. It is highly affected by outliers.

Example: For 10, 20, 30, 40, 50, range = $50 - 10 = 40$.

Outlier effect: For 10, 20, 30, 40, 500, range = $500 - 10 = 490$.

$$\text{Coefficient of Range} = (\text{Max} - \text{Min}) / (\text{Max} + \text{Min})$$

6.2 Interquartile Range and Quartile Deviation

$$IQR = Q_3 - Q_1$$

$$\text{Quartile Deviation} = (Q_3 - Q_1) / 2$$

$$\text{Coefficient of QD} = (Q_3 - Q_1) / (Q_3 + Q_1)$$

IQR measures the spread of the middle 50% of data. It is useful when outliers are present.

$$\text{Outlier rule: } X < Q_1 - 1.5(IQR) \text{ or } X > Q_3 + 1.5(IQR)$$

6.3 Mean Deviation

Mean deviation is the average absolute distance of values from mean or median.

$$\text{MD about mean} = (1/N) \sum |x - \bar{x}|$$

$$\text{MD about median} = (1/N) \sum |x - M|$$

For frequency distribution:

$$\text{MD about mean} = (1/N) \sum f |x - \bar{x}|$$

$$\text{MD about median} = (1/N) \sum f |x - M|$$

Mean deviation uses absolute values, so negative and positive deviations do not cancel.

6.4 Variance

Variance is the average of squared deviations from the mean. It combines all values in the dataset to measure spread.

$$\text{Population variance: } \sigma^2 = (1/N) \sum (x - \bar{x})^2$$

$$\text{Sample variance: } s^2 = [1/(n - 1)] \sum (x - \bar{x})^2$$

Variance is in squared units. For example, if the data is in marks, variance is in marks squared.

Variance example

Find variance of dataset 10, 60, 50, 30, 40, 20.

$$\text{Mean} = (10 + 60 + 50 + 30 + 40 + 20) / 6 = 35$$

$$\text{Variance} = [(10-35)^2 + (60-35)^2 + (50-35)^2 + (30-35)^2 + (40-35)^2 + (20-35)^2] / 6$$

$$\text{Variance} = 1750 / 6 = 291.7$$

6.5 Standard Deviation

Standard deviation is the square root of variance. It is easier to interpret than variance because it has the same unit as the original data.

$$\text{SD} = \sqrt{\text{Variance}}$$

$$\sigma = \sqrt{[(1/N) \sum (x - \bar{x})^2]}$$

SD value	Meaning
Small SD	Values are close to the mean. Data is more consistent.
Large SD	Values are far from the mean. Data has high variability.

Example: 48, 49, 50, 51, 52 has low SD. 10, 30, 50, 70, 90 has high SD even if the mean may be similar.

Relations between deviations

$$\text{SD} = (3/2) \times \text{QD}$$

$$\text{SD} = (5/4) \times \text{MD}$$

6.6 Coefficient of Variation

Coefficient of variation is a standardized measure of dispersion. It is useful for comparing variability of two datasets with different means or units.

$$CV = (SD / \text{Mean}) \times 100$$

Lower CV means more consistency. Higher CV means more variability.

Dataset	Mean	SD	CV	Conclusion
A	50000	5000	10%	More consistent
B	80000	20000	25%	More variable

7. Measures of Shape

Measures of shape describe the appearance of a distribution. The most important measures are skewness and kurtosis.

7.1 Skewness

Skewness measures the degree and direction of asymmetry in a distribution. It tells whether data is tilted to the left or right.

Type	Tail direction	Relation	Example
Positive skew / Right skew	Long tail on right	Mean > Median > Mode	Income, house prices
Negative skew / Left skew	Long tail on left	Mean < Median < Mode	Easy exam marks
Zero skew / Symmetric	Balanced tails	Mean = Median = Mode	Normal distribution

Dispersion tells the amount of variation. Skewness tells the direction of variation from symmetry.

Tests of skewness

- Mean, median and mode do not coincide.
- The graph is not a normal bell-shaped curve.
- Quartiles are not equidistant from the median.

Measures of skewness

$$\text{Pearson first coefficient} = (\text{Mean} - \text{Mode}) / SD$$

$$\text{Pearson second coefficient} = 3(\text{Mean} - \text{Median}) / SD$$

$$\text{Bowley coefficient} = (Q3 + Q1 - 2Q2) / (Q3 - Q1)$$

$$\text{Kelly percentile coefficient} = (P90 + P10 - 2\text{Median}) / (P90 - P10)$$

$$\text{Kelly decile coefficient} = (D9 + D1 - 2\text{Median}) / (D9 - D1)$$

Pearson coefficient is based on mean, median, mode and SD. Bowley coefficient is based on quartiles. Kelly coefficient is based on deciles or percentiles equidistant from median.

7.2 Kurtosis

Kurtosis describes the peak and tails of a distribution compared with a normal distribution. It helps understand how often outliers occur and how peaked or flat the curve is.

$$\text{Excess kurtosis} = \beta_2 - 3$$

Type	Condition	Shape	Outliers
Leptokurtic	$\beta_2 > 3$, excess kurtosis > 0	More peaked, heavy tails	High
Mesokurtic	$\beta_2 = 3$, excess kurtosis $= 0$	Normal distribution shape	Medium
Platykurtic	$\beta_2 < 3$, excess kurtosis < 0	Flatter peak, light tails	Low

Tails are the tapering ends on either side of a distribution. Heavy tails usually mean higher chance of outliers.

8. Data Visualization in Descriptive Statistics

Chart	Use
Histogram	Shows frequency distribution of numerical data. Useful for checking shape, spread and skewness.
Box Plot	Shows median, quartiles, IQR and outliers. Very useful for skewed data.
Bar Chart	Used for categorical data counts.
Pie Chart	Shows parts of a whole for categorical data.
Scatter Plot	Shows relationship between two numerical variables.

9. Complete Formula Sheet

Concept	Formula
Mean raw data	$\bar{x} = (\sum x) / N$
Mean frequency data	$\bar{x} = (\sum f x) / (\sum f)$
Shortcut mean	$\bar{x} = A + (\sum f d) / N$
Step deviation mean	$\bar{x} = A + [(\sum f u) / (\sum f)] \times i$
Geometric mean	$GM = \text{nth root of } (x_1 \times x_2 \times \dots \times x_n)$
Harmonic mean	$HM = N / \sum(1/x)$
Weighted mean	$\bar{x}_w = (\sum w x) / (\sum w)$
Grouped median	$\text{Median} = l + [((N/2) - C) / f] \times h$
Grouped mode	$\text{Mode} = l + [(f_1 - f_0) / (2f_1 - f_0 - f_2)] \times h$
Empirical relation	$\text{Mode} = 3\text{Median} - 2\text{Mean}$
Range	$\text{Max} - \text{Min}$
Coefficient of range	$(\text{Max} - \text{Min}) / (\text{Max} + \text{Min})$
IQR	$Q_3 - Q_1$
Quartile deviation	$(Q_3 - Q_1) / 2$

Concept	Formula
Coefficient of QD	$(Q3 - Q1)/(Q3 + Q1)$
Mean deviation about mean	$(1/N) \sum x - \bar{x} $
Variance	$(1/N) \sum (x - \bar{x})^2$
Standard deviation	$\sqrt{\text{Variance}}$
Coefficient of variation	$(SD / \text{Mean}) \times 100$
Pearson skewness 1	$(\text{Mean} - \text{Mode}) / SD$
Pearson skewness 2	$3(\text{Mean} - \text{Median}) / SD$
Bowley skewness	$(Q3 + Q1 - 2Q2)/(Q3 - Q1)$
Excess kurtosis	$\beta_2 - 3$

10. Interview and Exam Revision

Most important differences

Question	Answer
Mean vs Median	Mean is average and affected by outliers. Median is middle value and robust to outliers.
Variance vs SD	Variance is average squared deviation. SD is square root of variance and has original unit.
Range vs IQR	Range uses extremes and is outlier-sensitive. IQR uses middle 50% and is robust.
Skewness vs Kurtosis	Skewness measures asymmetry. Kurtosis measures peak and tails.
Descriptive vs Inferential	Descriptive summarizes given data. Inferential makes population-level conclusions from sample data.
Primary vs Secondary data	Primary is collected directly. Secondary is already collected data.

When to use what

Situation	Best choice
Clean numerical symmetric data	Mean and Standard Deviation
Outliers or skewed numerical data	Median and IQR
Categorical data	Mode and bar chart
Compare variability across datasets	Coefficient of Variation
Detect outliers	Box plot and IQR rule
Understand distribution shape	Histogram, skewness and kurtosis

11. Practice Questions

Try solving these questions after reading the notes.

- Find mean, median and mode for: 5, 7, 9, 12, 15, 18, 20.
- For data 3, 5, 7, 9, 15, 18, 20, 30, calculate range, Q1, Q3, IQR and quartile deviation.

- Find variance and standard deviation for: 10, 60, 50, 30, 40, 20.
- A class has 30 females with average 60 and 70 males with average 72. Find combined mean.
- Explain why median is better than mean when salary data has outliers.
- Differentiate positive skewness, negative skewness and zero skewness with one example each.
- Explain leptokurtic, mesokurtic and platykurtic distributions.
- When should you use coefficient of variation? Give one example.

12. Final Quick Summary

Topic	Core idea
Central tendency	Find the center of data using mean, median and mode.
Position	Find location of data using quartiles, deciles and percentiles.
Dispersion	Find spread using range, IQR, variance, SD and CV.
Skewness	Find direction of asymmetry.
Kurtosis	Find peak and tail behavior, especially outlier tendency.
Visualization	Use charts to understand patterns before modeling.

Final Golden Rule: When data has outliers, use Median, IQR and Coefficient of QD. When data is clean and symmetric, Mean, Range and Standard Deviation are effective.